Spike-count streaming across input rates

exp082 · 10 August 2026 · Complete

1. Abstract

We tested whether a spike-count readout trained across varying input rates could classify streamed digits when input rate and presentation time changed. Three frozen pyramidal–interneuron gamma (PING) networks were evaluated across eleven rates and four matched presentation and readout durations. With 200 ms available, mean accuracy rose from 26% at 0.5 Hz to 87.3% at 7.5 Hz and remained at 88.5% at 25 Hz. Across rates, shortening the window from 200 to 25 ms reduced mean accuracy from 69.1% to 36.5%. Sparse and brief inputs often failed to provide enough output spikes. The study supports broad rate tolerance within the tested protocol, but it does not isolate presentation time from evidence-accumulation time or establish a causal role for gamma rhythms.

In simple terms, the networks read digits reliably across many input strengths when given enough time and spikes, but struggled when the evidence was too sparse or brief.

2. Prospective design and scope

Three PING networks were trained independently across maximum-pixel Poisson rates from 0.5 to 25 Hz. Before evaluation, we froze their recurrent weights, output projections, and validation-selected checkpoints. Each decision uses only the output-LIF spikes emitted during that digit. The output neurons and counters reset at digit boundaries, while hidden PING state continues within each five-digit stream.

The leading account predicts useful decisions across much of the trained rate range, provided that enough output spikes arrive. A narrow rate optimum is the main rival. Two measurement effects could also shape the result: longer integration could mimic rate tolerance, and silent output windows could dominate failure. The fixed-200-ms psychometric screens rate tolerance. The duration–rate grid shows how performance changes as matched windows shorten, but it cannot separate presentation time from readout time.

The primary estimand is condition-level accuracy,

$$A_{d,r,s} = \frac{1}{N}\sum_{i=1}^{N} 1_{\hat{y}_{i,d,r,s}=y_i},$$

where $A_{d,r,s}$ is accuracy for duration d, rate r, and trained seed s. Each cell contains $N =$ 200 digits; $\hat{y}$ is the predicted label and y is the true label. Output silence is a companion diagnostic. We summarize the three seed-level accuracies rather than treating 600 presentations as independent network replicates.

Chance-level accuracy throughout the rate range would falsify useful transfer of the trained readout. A narrow optimum would contradict broad rate tolerance. Persistent silence at moderate rates would instead identify output activity as the limiting failure mode. The duration grid tests robustness, but cannot distinguish less presentation time from less integration time.

> The experiment and publication scaffold were committed before execution. They froze the networks, checkpoint policy, readout, rate and duration grid, sampling scale, and planned outputs. The historical plan did not pre-register directional thresholds or an uncertainty model. Those omissions limit confirmatory interpretation, but the executed protocol itself is prospective.

All 132 seed-level cells completed: 26,400 classifications through three pretrained networks, with no retraining or parameter search. Execution matched the frozen rate grid, duration grid, sampling scale, and checkpoint policy. The study uses validation-selected classifiers and held-out streams. It neither isolates training distribution from readout nor includes a fixed-rate control. Three training seeds constrain population-level inference.

3. Investigations

3.1 How output spikes become a decision

The first investigation shows how the trained readout forms a decision. At timestep t, the excitatory spike vector $\boldsymbol{s}^{E(t)}$ drives ten learned output-LIF units. For a presentation beginning at a and ending before b, class c accumulates

$$z_{c(u)} = \sum_{t=a}^{u} s_c^{\mathrm{out}(t)}, \quad a \leq u < b,$$

where $s_c^{\mathrm{out}(t)}$ is the spike emitted by output unit c. The final prediction is $\arg\max_c z_{c(b-1)}$. If rhythmic bursts deliver useful evidence packets, classes can exchange the lead before one builds a stable margin. Sparse failure should instead produce few or no output spikes.

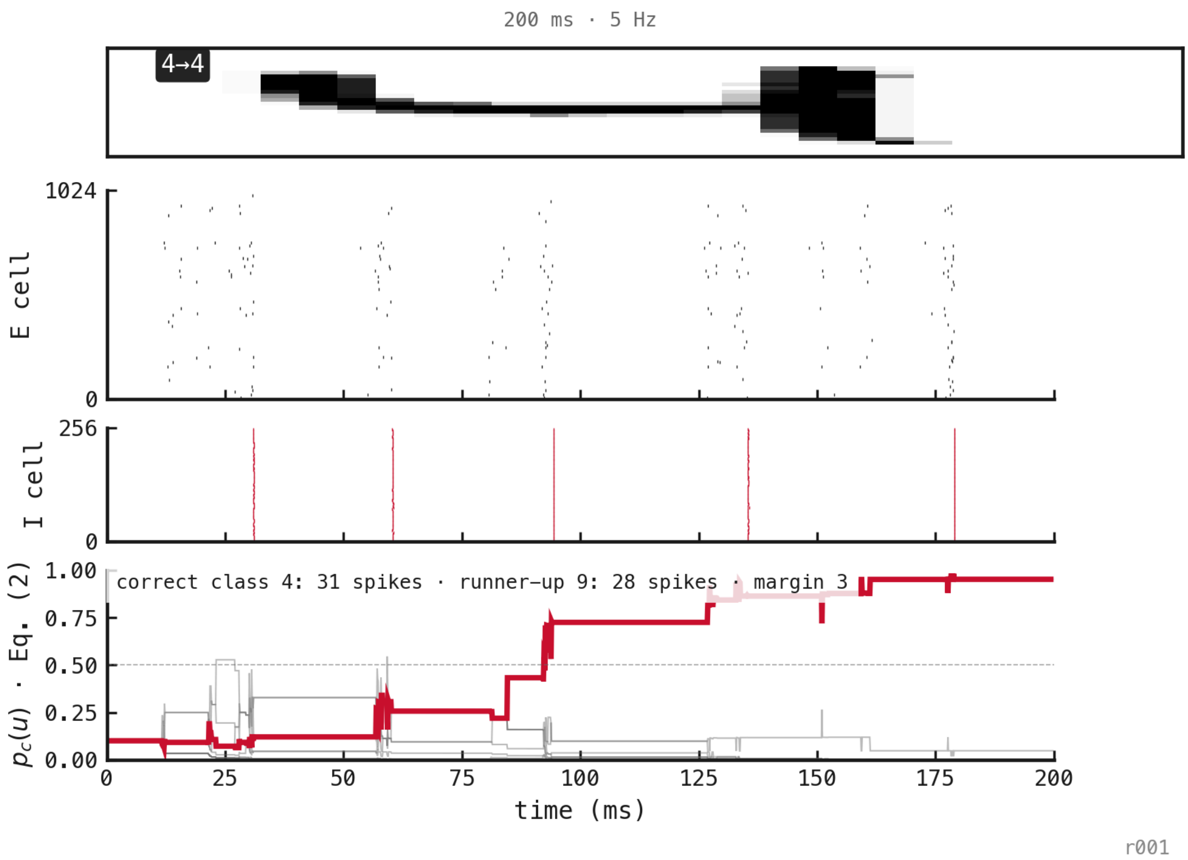

Figure 1: A correctly classified 200 ms presentation at 5 Hz, selected as the first success in a pre-existing matched stream. The figure explains the readout; it does not estimate accuracy. Red marks the true and winning class.

In this digit-4 presentation, output spikes increment class counts around successive population bursts. The display applies $p_{c(u)} = \mathrm{softmax}_{c(\boldsymbol{z}(u))}$ to those counts. It is not a calibrated posterior. Because $\frac{p_{c(u)}}{p_{j(u)}} = \exp\big(z_{c(u)} - z_{j(u)}\big)$, a small integer margin can look decisive. The counts determine the winner; softmax only displays their relative separation.

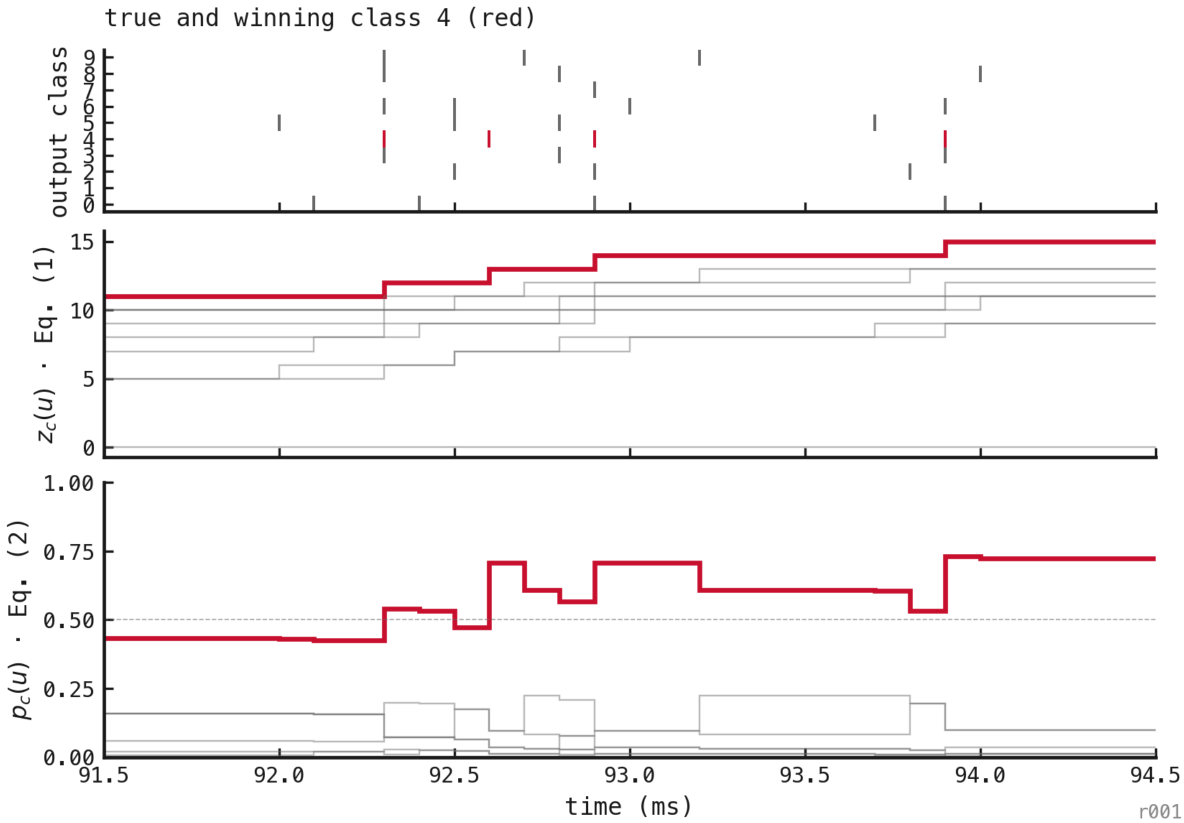

Figure 2: A post-hoc enlargement of 91.5–94.5 ms in the same trial. Each spike increments one cumulative count. Softmax normalization can lower another class's displayed share even when its count does not change.

The trajectory is consistent with PING bursts delivering packets of class evidence. One outcome-selected example cannot show that each cycle improves the decision, or that rhythmic packaging causes the aggregate pattern.

3.2 Classification in a changing stream

The second investigation changes input rate and presentation duration at digit boundaries while enforcing $T_{\text{readout}} = T_{\text{presentation}}$. Boundary resets should prevent class evidence from leaking across labels, although continuing hidden state can shape early activity. Five registered conditions span sparse, dense, short, and matched presentations. This stream is a qualitative stress test, not an accuracy estimate.

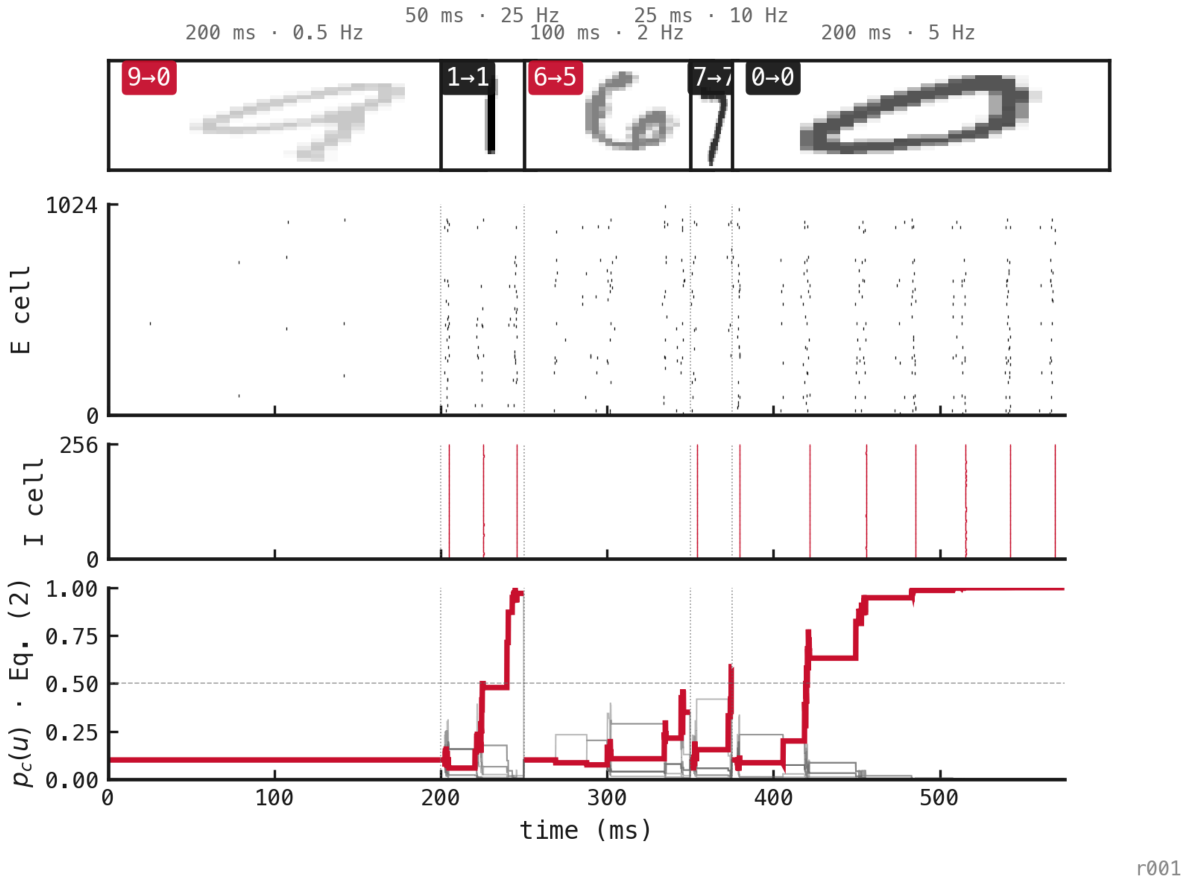

Figure 3: A five-digit stream classified by the native output-LIF spike-count readout. Counts reset at each boundary; hidden PING state continues. Thumbnail opacity increases with input rate, and badges show the true and predicted labels.

Three of five presentations were correct. The 0.5 Hz presentation produced no output spikes and failed; the 2 Hz presentation also failed. Presentations at 5, 10, and 25 Hz were correct despite lasting from 25 to 200 ms. The stream verifies matched-window operation and exposes a silent sparse-input failure. Five outcomes cannot estimate reliability.

3.3 Rate and duration robustness

The factorial evaluation crosses four matched presentation/readout durations with all eleven training rates. Each condition contains 200 digits for each network. The leading account predicts that accuracy will rise as rate and duration supply more output events, then remain useful across a broad upper-rate region.

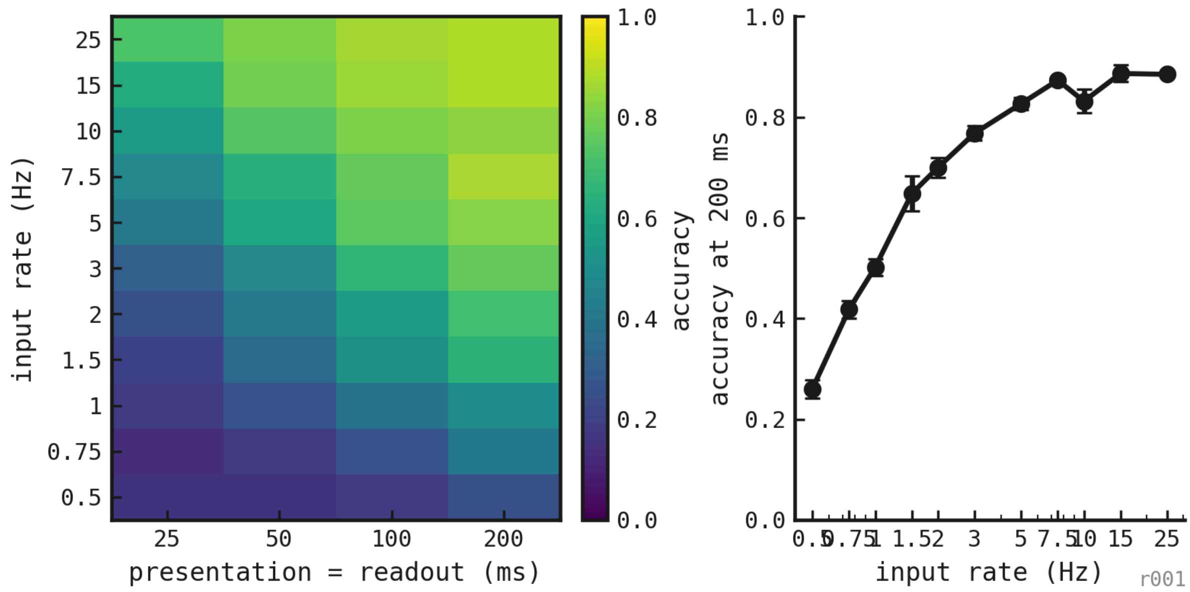

Figure 4: Held-out accuracy across matched presentation/readout durations and trained input rates. The right panel fixes duration at 200 ms to separate rate tolerance from integration time.

Mean accuracy increased monotonically with duration: 36.5% at 25 ms, 49% at 50 ms, 60.4% at 100 ms, and 69.1% at 200 ms. Mean output silence fell from 31.1% to 4.8%. Shorter presentations leave less time for both hidden dynamics and evidence accumulation. This design cannot separate those effects.

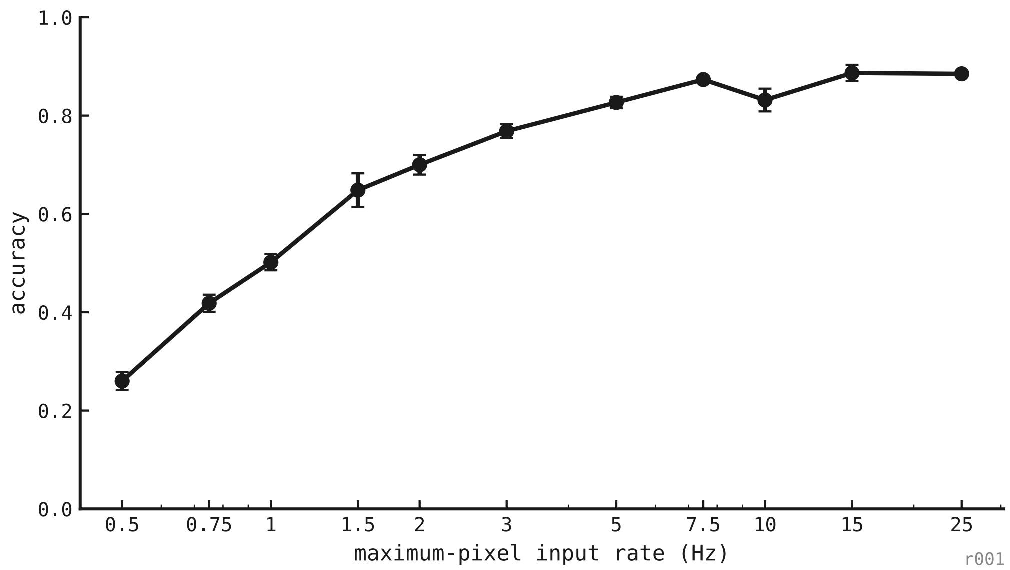

Figure 5: Input-rate psychometric with presentation and spike-count window fixed at 200 ms. Each rate contains 200 held-out digit presentations for each of three trained seeds.

At 200 ms, mean accuracy rose from 26% at 0.5 Hz to 76.8% at 3 Hz and 87.3% at 7.5 Hz. Across 10–25 Hz, it remained between 83.2% and 88.7%. The dense end did not collapse. At 15 Hz, seed-level accuracy ranged from 87% to 92%. The sparse edge was weaker: at 0.5 Hz, accuracy ranged from 23.5% to 29.5%, with 37.2% silent windows on average.

4. Executed methods

4.1 Networks, data, and frozen selection

Three PING networks were trained independently on MNIST. Each image presentation sampled one of the eleven registered input rates. A learned projection connected 1024 excitatory cells to ten output-LIF class units, whose spike counts served as logits. Validation accuracy selected one checkpoint per seed. All weights and checkpoints remained fixed during the study.

4.2 Streaming protocol

Input pixels generated independent Bernoulli spikes at 0.1 ms resolution. Each evaluation cell comprised 40 independent five-digit streams per seed, giving 200 decisions. We simulated five streams per batch, with separate neuronal state for each stream. Hidden state continued only within a stream. At every digit boundary, the output-LIF state and counter reset. Each decision therefore used its matched presentation window alone.

4.3 Factorial evaluation and diagnostics

The factorial evaluation crossed presentations of 25, 50, 100, and 200 ms with input rates from 0.5 to 25 Hz. The fixed-duration psychometric is the 200 ms slice. For each seed and condition, we retained accuracy, output spikes per presentation, silent-window fraction, and excitatory and inhibitory population rates. The plots summarize seed-level accuracy; they do not pool seeds into one nominal replicate.

4.4 Illustrative trajectories

The variable stream used five fixed duration–rate pairs and supports only qualitative interpretation. The single-trial figure shows the first correct presentation in a pre-existing 200 ms, 5 Hz stream. We selected it to explain a successful readout. The 91.5–94.5 ms enlargement was chosen post hoc around a conspicuous transition. Neither selection contributes to aggregate accuracy.

4.5 Deviations, uncertainty, and robustness limits

All factorial cells completed without a known scientific deviation from the frozen design. The study lacks population-level intervals, additional training seeds, an independent stream-bank repeat, and adjusted condition comparisons. Agreement across three networks supports robustness within the tested protocol, not across a broader network population. Because validation selected the checkpoints, this is a deployment evaluation rather than an account of final-epoch dynamics.

5. Conclusion

Across these three networks, the native matched-window readout worked over a broad part of the trained rate range. At 200 ms, accuracy remained high from moderate through dense rates, weakening the narrow-optimum rival. Longer windows improved accuracy and reduced silence. The weak 0.5 Hz edge points to insufficient output evidence as the sparse-input failure mode.

The study does not explain why variable-rate training works, establish a causal role for PING cycles, or show superiority over the exp048 mean-voltage decoder. That comparison changed both training distribution and readout. A stronger successor should compare fixed-rate and variable-rate training under matched readouts. It should add training seeds, predefine uncertainty and decision thresholds, and manipulate presentation time separately from integration time.